

3(a)(i)	$F = kx$	C1
	$F_1 = 800 \times 0.045$ $= 36 \text{ N}$	A1
3(a)(ii)	$(E =) \frac{1}{2}kx^2$ or $\frac{1}{2}Fx$ or area under graph	C1
	$\frac{1}{2} \times 800 \times (0.045)^2$ or $\frac{1}{2} \times 36 \times 0.045 = 0.81 \text{ (J)}$	A1
3(b)(i)	efficiency = $(0.72 / 0.81) \times 100$ $= 89\%$	A1
3(b)(ii)	$E = \frac{1}{2}mv^2$	C1
	$p = mv$	C1
	$0.72 = \frac{1}{2} \times 0.020 \times v^2$ and $p = 0.020 \times v$ $p = 0.17 \text{ N s}$	A1
3(c)(i)	$(\Delta)E = mg(\Delta)h$	C1
	$h = 0.60 / (0.020 \times 9.81) = 3.1 \text{ m}$	A1
3(c)(ii)	$F = (0.72 - 0.60) / 3.1$	C1
	$= 0.039 \text{ N}$	A1
3(c)(iii)	resultant force on ball is less (than that with air resistance) so time (taken) is more (than T)	B1